# Feasibility Memo: AI-Assisted Triage Modelling

## One-sentence verdict

The dataset is suitable to build and internally validate a **first baseline** triage-acuity model, but it is a **US-derived, adult-only baseline**, not a Caribbean-ready deployment — the underlying data quality is good, but the population, insurance categories, and race/ethnicity balance do not yet represent our intended deployment context.

## Dataset summary

**Size and structure.** The cleaned extract contains **55,121 emergency department encounters across 225 columns** (the raw export has 226 columns before an unnamed row-index column is dropped on load). Of the 225 columns, **25 are structured clinical, demographic, and administrative fields**, and the remaining **200 are chief-complaint flags** (`cc_*`), each a 0/1 indicator of whether that complaint was recorded for the encounter. The dataset is therefore best described as **wide and sparse rather than deep** most `cc_*` cells are 0 for any given patient.

**Clinical variables available.**

- **Vitals recorded at triage:** heart rate, systolic and diastolic blood pressure, respiratory rate, oxygen saturation, an oxygen-device flag (room air vs supplemental oxygen), temperature (recorded in **Fahrenheit**, not Celsius), and point-of-care glucose.
- **Demographics:** age, gender, ethnicity, race, language, religion, marital status, employment status, and insurance status. Race, ethnicity, and insurance status are flagged as fairness-sensitive throughout the project.
- **Administrative/arrival fields:** department, arrival mode, arrival month/day/hour band.
- **Chief complaints:** 200 binary flags, which collapse into **12 clinically meaningful body systems** (see below).
- **Excluded as leakage:** `disposition` and `previousdispo`. Both describe what happened to the patient *hours after* triage, once workup was under way, so neither is ever used as a model input for a triage-time prediction.

**The target.** `esi` (the Emergency Severity Index, our Triage_Level) is an ordinal 1–5 scale assigned at the door, where 1 is most urgent. It has **zero missing labels** across all 55,121 encounters. The class balance is:

| ESI level | Meaning | Encounters | % of sample |
|---|---|---|---|
| 1 | Resuscitation | 77 | 0.1% |
| 2 | Emergent | 17,924 | 32.5% |
| 3 | Urgent | 27,010 | 49.0% |
| 4 | Less urgent | 8,896 | 16.1% |
| 5 | Non-urgent | 1,214 | 2.2% |

This is a realistic shape for a hospital ED —dominated by ESI 2–3  but it also means ESI 1 is the rarest and hardest class to learn reliably.

**Cleaning and preprocessing performed.** Before any modelling, the pipeline:

1. Would drop any encounter with no `esi` label (none needed dropping see finding below).
2. Coerced the vitals and age to numeric values.
3. Flagged physiologically impossible readings (e.g. a heart rate outside 20–250 bpm, or a systolic BP outside 50–300 mmHg) as missing rather than capping them.
4. Filled the resulting gaps: vitals with the column median; the oxygen-device flag and chief-complaint flags with 0 ("not recorded"); blank demographic/administrative categories with the explicit label "Unknown".
5. Rounded `esi` to a whole number.

**A key finding from profiling:** the raw structured columns arrived with **zero missing values** across all 25 fields  the source extract was already pre-cleaned. The one real data-quality issue our own outlier check surfaced was a **small number of physiologically implausible readings**: 4 respiratory-rate values and 25 glucose values fell outside plausible bounds and were corrected via the median-imputation step above. Every other vital had zero implausible readings. Separately, ordinary statistical (IQR-fence) outliers were common and were **retained, not removed** ranging from 578 flagged heart-rate readings up to 5,648 flagged glucose readings — because in an ED population, real physiological extremes are exactly the encounters most clinically important to keep.

**Relationship with the target.** Correlating each vital against `esi` gives: oxygen saturation **r = 0.178** (positive — lower SpO2 tracks *higher* acuity, i.e. lower ESI), respiratory rate **r = −0.097**, heart rate **r = −0.095**, glucose **r = −0.078**, temperature **r = −0.022**, diastolic BP **r = 0.046**, and systolic BP **r = 0.001**. Age correlates at **r = −0.237** — the single strongest bivariate signal found in this profiling exercise and being on a supplemental-oxygen device at triage correlates at **r = −0.209**. Among chief complaints, the strongest associations with higher acuity are chest pain (**r**

**= −0.164**), shortness of breath (**r = −0.150**), suicidal ideation (**r = −0.143**), and alcohol intoxication (**r = −0.142**).

**Chief-complaint structure.** The 200 flags sort into 12 body systems; only one flag never fires in this sample. Complaint volume concentrates rather than scatters evenly: Trauma/Injury/MSK accounts for 13,094 flag-hits, Gastrointestinal 10,542, Respiratory 6,693, and so on down to Endocrine/Metabolic/Heme at 875. The single most common individual complaints are abdominal pain (6,717 encounters), an "other" catch-all (4,491), chest pain (3,712), and shortness of breath (3,098).

A missingness map, the ESI/age distribution, the race/ethnicity breakdown, the top chief complaints, vitals split by ESI level, and the vitals-plus-ESI correlation matrix were generated directly from this dataset and are available in `figs/` in the dashboard notebook: missingness before cleaning (**Plot 1**), ESI class balance and age distribution (**Plot 2**), race/ethnicity breakdown (**Plot 3**), top chief complaints (**Plot 4**), vitals by ESI level (**Plot 5**), and the vitals-plus-ESI correlation matrix (**Plot 6**).

# Top 3 quality concerns

**1. A small number of physiologically implausible vitals required correction.** Even though the raw structured data was otherwise complete, our outlier check found 4 respiratory-rate readings and 25 glucose readings outside plausible physiological bounds (e.g. an implausibly low or high value that cannot reflect a real patient). This matters for AI triage because an uncorrected implausible value  a glucose reading in the thousands, for instance would distort a learned threshold and could push a model to make a confidently wrong call on exactly the encounters where correctness matters most. The correction step (flag as missing, then median-impute) is small in volume but must be repeated on every future data pull, since real-world streams will rarely arrive this tidy.

**2. Statistical outliers are common and vary sharply by vital, and must not be mistaken for errors.** IQR-fence outlier counts ranged from 578 (heart rate) to 5,648 (glucose) all retained as real values, not corrected. This matters for AI triage because a pipeline that automatically strips "outlier" rows would disproportionately remove the sickest patients, the ones a triage model most needs to learn from. Outlier handling must keep distinguishing statistical rarity (a genuinely high heart rate) from clinical implausibility (a heart rate no living patient could have).

**3. Demographic and geographic representativeness does not match our deployment context.** The race distribution is dominated by White/Caucasian (29,435) and Black/African American (15,963) patients, with much smaller Asian (175), American Indian/Alaska Native (66), and Native Hawaiian/Pacific Islander (20) groups; ethnicity is 82% Non-Hispanic; and insurance status uses US-specific categories (Medicaid, Medicare, Commercial, Self-pay) with no equivalent to a Caribbean health-financing system. This matters for AI triage because small subgroups make it statistically hard to check whether a model performs equitably for them, and because the sample's disease mix, arrival patterns, and insurance categories may simply not resemble a Caribbean ED population.

# Top 3 reasons to proceed

**1. The target is clean and fully populated.** `esi` has no missing labels across all 55,121 encounters and sits entirely on a well-defined, clinically standard 1–5 scale, giving a real and learnable target rather than one we would need to construct or approximate.

**2. The vitals plausibly separate acuity, in clinically expected directions.** Oxygen saturation shows the strongest single-vital correlation with `esi` (r = 0.178), and respiratory rate and heart rate both correlate around r = −0.10 (−0.097 and −0.095 respectively) consistent with respiratory rate's reputation as the earliest deterioration signal. The box-plot comparison of vitals across ESI levels (**Plot 5**) shows this separation visually rather than only numerically.

**3. The chief complaints collapse into a manageable, clinically sensible structure.** Despite 200 raw flags, they sort cleanly into 12 body systems with essentially no leftover complaints, and complaint volume concentrates in a few systems  Trauma/Injury/MSK, Gastrointestinal, and Respiratory rather than spreading thinly and uniformly. Individual complaint-to-acuity correlations are clinically sensible (chest pain and shortness of breath rank highest), which makes this a tractable feature-engineering task rather than an unworkable mass of sparse columns (**Plot 4**).

# Caveats

- **Retrospective extract; site structure not fully documented.** This is historical US academic-hospital data. The `dep_name` field records three coded values (A, B, C) none of the three notebooks state whether these are departments within one hospital or separate sites, so "single-site" should be confirmed with the original data source rather than assumed. Either way, it supports building a baseline, not prospectively validating clinical performance.
- **Adult-only scope.** Ages run from 18 to 107 (mean 55.3); there are no paediatric encounters. Any deployment involving children would need separate data and separate validation.
- **Bias and equity monitoring is ongoing, not a one-off.** Race, ethnicity, and insurance status (a proxy for access) are the columns most likely to encode inequity into a triage model. Small subgroup sizes (e.g. Asian n = 175, Native Hawaiian/Pacific Islander n = 20) make equity checks statistically fragile, so the ESI distribution within each race category should remain a standing check, not a single plot produced once.
- **Leakage is excluded, and that is itself a limitation on realism.** `disposition` and `previousdispo` are outcomes decided hours after triage and are never used as inputs — correct for a genuine triage-time model, but it means this analysis says nothing yet about downstream outcomes such as admission.
- **Validation is still bivariate, not model performance.** The correlations reported here (maximum single-vital |r| ≈ 0.18, maximum single-complaint |r| ≈ 0.16, age at r = −0.237)

are directional sanity checks on a real label, not evidence of predictive accuracy. No train/test split, calibration check, or clinical-trial-style validation has been run yet.

- **Clinical oversight remains essential, especially at the extremes.** ESI 1 (Resuscitation) is only 77 of 55,121 encounters (0.1%)  the rarest and most safety-critical class. Any model built on this data needs a human-in-the-loop safety net, particularly for anything flagged as high acuity.
- **Imputation choices are defensible but not neutral.** Median-imputing the small number of implausible vitals, and filling blank categories with "Unknown", are reasonable defaults but they are choices, and this memo states them plainly rather than implying the data needed no judgement calls at all.
- **Units matter.** Temperature is recorded in Fahrenheit throughout. Any downstream chart, threshold, or clinician-facing display must say so explicitly, since a normal reading would otherwise misread as hypothermia.

# Recommendation to the ED Board

**Proceed to build and internally validate a baseline ESI-prediction model on this dataset**, on the explicit understanding that it is a development exercise on US data, not a Caribbean-ready clinical tool. Before any patient-facing pilot is considered, we recommend the Board require:

1. **A locally sourced validation cohort** from a Caribbean ED, however small, before any claim of transferability is made.
2. **A standing equity check** the ESI distribution by race, ethnicity, and insurance status reviewed at each project milestone, not only at kickoff.
3. **Human-in-the-loop oversight**, with mandatory clinician review of any case the model places at ESI 1–2, given how rare and safety-critical that class is in the training data.
4. **Explicit documentation of unit and category assumptions** (Fahrenheit temperature, US arrival-mode and insurance categories) attached to any model artefact, so nobody downstream inherits these assumptions silently.

# Top-10 Feature Shortlist — ESI (Triage_Level) Prediction

All features below exist as-recorded in the cleaned extract (`triage_cleaned_v1.csv`). Ranking is by strength of the bivariate correlation with `esi` found during profiling, combined with a one-sentence clinical justification for why a triage nurse would actually rely on that signal. Note that `esi` runs 1 (most urgent) to 5 (least urgent), so a **negative** correlation means the feature rises as urgency increases.

| Rank | Feature | Why it is predictive |
|---|---|---|
| 1 | `age` | Strongest bivariate signal found (r = −0.237 with `esi`); clinically, older patients have less physiological reserve, so the same insult tends to be triaged more urgently. |
| 2 | `triage_vital_o2_device` | Second-strongest signal (r = −0.209); already needing supplemental oxygen at the point of triage is itself a direct marker of acute respiratory compromise, independent of the SpO2 number recorded alongside it. |
| 3 | `triage_vital_o2` (oxygen saturation) | Strongest single vital-sign correlation (r = 0.178, positive because higher SpO2 tracks lower acuity); hypoxia is an explicit ESI "danger-zone vital" that can up-triage a patient on its own. |
| 4 | `cc_chestpain` | Strongest chief-complaint correlation (r = −0.164); chest pain with risk factors is a canonical ESI-2 trigger for a possible cardiac or vascular emergency. |
| 5 | `cc_shortnessofbreath` | Second-strongest complaint correlation (r = −0.150); respiratory distress is one of the explicit "can't-wait" criteria in the ESI algorithm. |

| | | |
|---|---|---|
| 6 | `cc_suicidal` | r = −0.143; psychiatric emergencies involving risk of self-harm are explicitly high-acuity under ESI, regardless of physical vitals. |
| 7 | `cc_alcoholintoxication` | r = −0.142; intoxication carries real risk of airway compromise and fluctuating consciousness, both of which push a nurse toward a lower (more urgent) ESI level. |
| 8 | `cc_alteredmentalstatus` | r = −0.132; altered mentation is a direct ESI-1/2 trigger, since it can signal anything from stroke to sepsis to overdose. |
| 9 | `triage_vital_rr` (respiratory rate) | r = −0.097, the strongest of the two; clinically regarded as the earliest sign of patient deterioration. |
| 10 | `triage_vital_hr` (heart rate) | r = −0.095, essentially matching respiratory rate; tachycardia is a recognised stress/shock signal and sits on the ESI danger-zone vitals list alongside it. |

**Note on scope:** this shortlist deliberately favours features with both a measured correlation *and* a clear clinical mechanism, rather than simply the ten highest correlations in the dataset. Several `cc_*` flags with near-zero prevalence were excluded even where a correlation could be computed, since as flagged during profiling correlations on very sparse, low-count flags are statistically unstable and easy to over-read.

# Correlation Analysis — Chief Complaints vs Clinical Variables

This note extends the profiling and visualisation work in Tutorials 2–3 to the specific complaint-vital relationships the ED Board asked about. Correlations use the same point-biserial approach as `df[cols].corrwith(df[TARGET])` in Tutorial 2 — here applied directly between a 0/1 complaint flag and a numeric vital, over all 55,121 cleaned encounters in `triage_cleaned_v1.csv`. All figures below were computed directly from the dataset; none are estimated or assumed.

### Chest pain and heart rate (`cc_chestpain` vs `triage_vital_hr`)

- Mean heart rate with chest pain recorded (n = 3,712): **85.3 bpm**
- Mean heart rate without chest pain recorded (n = 51,409): **86.5 bpm**
- Correlation: **r = −0.018** (essentially no linear relationship)

**Clinical meaning:** chest pain in the ED does not, on its own, produce a reliably elevated heart rate in this sample many chest-pain presentations are haemodynamically stable at the point of triage, which is exactly why ESI relies on additional criteria (risk factors, ECG, troponin) rather than heart rate alone to escalate these patients.

**Limitation:** this is a single bivariate check. It does not rule out heart rate mattering in combination with other chest-pain-specific risk factors, which this dataset does not separately record (e.g. no ECG findings, no troponin field exists in the extract).

### Fever and temperature (`cc_fever*` flags vs `triage_vital_temp`)

The dataset records fever as four separate flags (`cc_fever`, `cc_fever-75yearsorolder`, `cc_fever-9weeksto74years`, `cc_feverimmunocompromised`); any one of these being positive was treated as "fever recorded".

- Mean temperature with a fever flag recorded (n = 1,126): **99.9 °F**
- Mean temperature without any fever flag (n = 53,995): **98.1 °F**
- Correlation: **r = 0.315** — the strongest relationship found in this analysis.

**Clinical meaning:** this is the expected result and the strongest signal in the set — patients whose chief complaint includes a fever flag really do run measurably hotter, which is reassuring evidence that the temperature field and the fever complaint flags are internally consistent and clinically usable together.

**Limitation:** temperature is recorded in **Fahrenheit** in this dataset; any downstream threshold (e.g. the standard 100.4 °F / 38 °C fever cut-off) must use the Fahrenheit value, not a Celsius assumption.

### Shortness of breath and oxygen saturation (`cc_shortnessofbreath` / related flags vs `triage_vital_o2`)

- Mean SpO2 with `cc_shortnessofbreath` recorded (n = 3,098): **95.6%**
- Mean SpO2 without it (n = 52,023): **97.1%**
- Correlation: **r = −0.161**
- Combining `cc_shortnessofbreath` with the related `cc_dyspnea`, `cc_breathingdifficulty`, and `cc_breathingproblem` flags (n = 3,822 with any of these recorded) gives a similar picture: mean SpO2 95.6% vs 97.1%, r = −0.183.

**Clinical meaning:** patients reporting breathing difficulty do run measurably lower oxygen saturations, consistent with respiratory distress complaints reflecting a real physiological drop in oxygenation, not just a symptom label.

**Limitation:** a 1.5-percentage-point mean difference is real but modest; SpO2 alone will miss patients compensating well at rest, so it should not be read as a substitute for full respiratory assessment.

### Trauma and blood pressure (`cc_fulltrauma` / `cc_modifiedtrauma` vs blood pressure)

- Encounters with either trauma flag recorded: **n = 20** (out of 55,121)
- Mean systolic BP: trauma 134.1 mmHg vs no trauma flag 133.7 mmHg (r = 0.000)
- Mean diastolic BP: trauma 76.8 mmHg vs no trauma flag 79.5 mmHg (r = −0.004)
- Mean heart rate: trauma 82.3 bpm vs no trauma flag 86.4 bpm (r = −0.005)

**Clinical meaning:** no meaningful relationship was found but this is very likely an artefact of how trauma is coded in this extract rather than a genuine clinical finding. Only 20 encounters carry the specific `cc_fulltrauma`/`cc_modifiedtrauma` flags; the great majority of trauma-type presentations in this dataset are instead spread across the much larger Trauma/Injury/MSK body-system group (49 flags, 13,094 total flag-hits), which is dominated by minor, extremity-specific injuries (ankle pain, arm injury, and similar) rather than major trauma with haemodynamic compromise.

**Limitation:** with n = 20, this specific check is too small to draw a reliable conclusion either way. A more informative version of this analysis would define "trauma" as the full Trauma/Injury/MSK group and would need to separate major trauma from minor extremity injury a distinction this extract does not make explicit.

### Abdominal pain and pain score

**This relationship cannot be reported as requested.** The cleaned extract contains no numeric pain-score field only chief-complaint flags such as `cc_abdominalpain`, `cc_abdominalcramping`, and `cc_abdominalpainpregnant`. As the closest available substitutes, we checked abdominal pain against `esi` and against heart rate directly:

- `cc_abdominalpain` vs `esi` (n = 6,717 with the flag): mean ESI 2.8 vs 2.9 without it; **r = −0.024**
- `cc_abdominalpain` vs `triage_vital_hr`: mean HR 87.1 bpm vs 86.4 bpm without it; **r = 0.015**

**Clinical meaning:** abdominal pain shows only a very weak tilt toward higher acuity and essentially no relationship with heart rate in this sample, which is clinically plausible abdominal pain spans everything from minor cramping to a surgical abdomen, so a single flag without a severity score should not be expected to separate acuity well.

**Limitation:** without a pain-severity field, this is the best available check, not the one that was asked for. If a pain-score field exists in the original raw source but was dropped before this extract, it would be worth requesting for a future iteration of this analysis.

## Summary table

| Relationship | Correlation | Direction consistent with clinical expectation? |
|---|---|---|
| Chest pain ↔ heart rate | r = −0.018 | No — expected but not found; ESI relies on other chest-pain criteria |
| Fever ↔ temperature | r = 0.315 | Yes — strongest relationship in this analysis |
| Shortness of breath ↔ SpO2 | r = −0.161 to −0.183 | Yes — lower SpO2 with breathing-related complaints |
| Trauma flags ↔ blood pressure | r ≈ 0.000 | Inconclusive — too few encounters (n = 20) carry these specific flags |
| Abdominal pain ↔ acuity/heart rate (no pain score available) | r = −0.024 / 0.015 | No — weak, plausible given no severity field exists |